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Paleomagnetic study of upper Miocene rocks from northern Chile: Implications for the origin of late Miocene-Recent tectonic rotations in the southern Central Andes

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Abstract. Paleomagnetic studies in the southern Central Andes have shown the widespread presence of clockwise vertical-axis rotations. Rock units sampled in these studies, however, are heterogeneously distributed in stratigraphic age with most paleomagnetic data from northern Chile being from Mesozoic and lower Tertiary rocks, whereas most data in the southern Altiplano, Puna, and Cordillera Oriental are from upper Tertiary rocks. In this paper we present the results of a paleomagnetic study on upper Miocene sedimentary rocks and ignimbrites from the Precordillera of northern Chile (at 22°S). These rocks are coeval with the initiation of crustal shortening in the eastern foreland fold-thrust belt which some tectonic models relate to oroclinal rotation of northern Chile. Primary magnetizations in rocks from widely distributed sites in two ignimbrites indicate that no relative rotations have occurred between sites, suggesting the study area has acted as a single coherent block with respect to vertical-axis rotational deformation. Although minor inadequate sampling of paleosecular variation can affect our data set, its time-averaged paleomagnetic direction is indistinguishable from the expected late Miocene reference direction indicating no paleomagnetically significant rotation of the study area since circa 11 Ma. This suggests that late Miocene-Recent oroclinal rotation of the northern Chilean forearc, if present, must be either very low or nonuniform. A direct implication of this result is that much of the unquestionable tectonic rotations detected from upper Miocene rocks in the southern Central Andes is of local origin. Available structural data permit us to relate several of these rotations with shear in dextral transfer zones in the foreland thrust belt on the east side of the Andes.

1. Introduction

Paleomagnetic studies have shown that vertical-axis rotations constitute an important component of the deformation in the southern Central Andes [e.g., *MacFadden et al.*, 1990; *Hartley et al.*, 1992; *Riley et al.*, 1993; *MacFadden et al.*, 1995; *Randall et al.*, 1996; *Aubry et al.*, 1996]. Most paleomagnetic data from northern Chile were obtained from Mesozoic and lower Tertiary rocks. In contrast, most paleomagnetic data from southern Bolivia and northwestern Argentina belong to upper Tertiary rocks (Figure 1a). Despite these differences in rock ages, the paleomagnetic data set indicates the systematic presence of clockwise rotations throughout the region, bringing about the formulation of several tectonic models.

Beck [1988] suggested that clockwise declination anomalies in northern Chile are the result of small-block rotations. Appealing to plate convergence models for the Nazca-South America pair, he proposed a mechanism that involves orogen-parallel dextral shear in response to oblique convergence. *Isacks* [1988] proposed that the continental margin was originally straighter and that northern Chile underwent a regional clockwise rotation during oroclinal bending of the Central Andean forearc, which was accommodated by late Cenozoic differential shortening in the eastern foreland thrust belt. A further proposal combining both the orogen-parallel shear and oroclinal bending mechanisms was presented by *Beck et al.* [1994], who showed that a late Cenozoic regional clockwise rotation of northern Chile must increase the convergence obliquity enhancing orogen-parallel shear. Other authors [*Hartley et al.*, 1988, 1992; *Forsythe and Chisholm*, 1994; *Randall et al.*, 1996] have linked some rotations in northern Chile with local structures, but the proposed rotation mechanisms are not widely applicable and do not explain the consistency of the rotation sense throughout the southern Central Andes.

Paleomagnetic data from southern Bolivia and northwestern Argentina have been interpreted as either evidence of oroclinal clockwise rotation of the southern Central Andes [*MacFadden et al.*, 1995; *Watts et al.*, 1995] or the result of local rotations induced by dextral shear [*Aubry et al.*, 1996; *Somoza et al.*, 1996].

An estimation of the magnitude, distribution, and nature of Neogene rotational deformation in northern Chile is difficult with the paucity of paleomagnetic data from upper Tertiary

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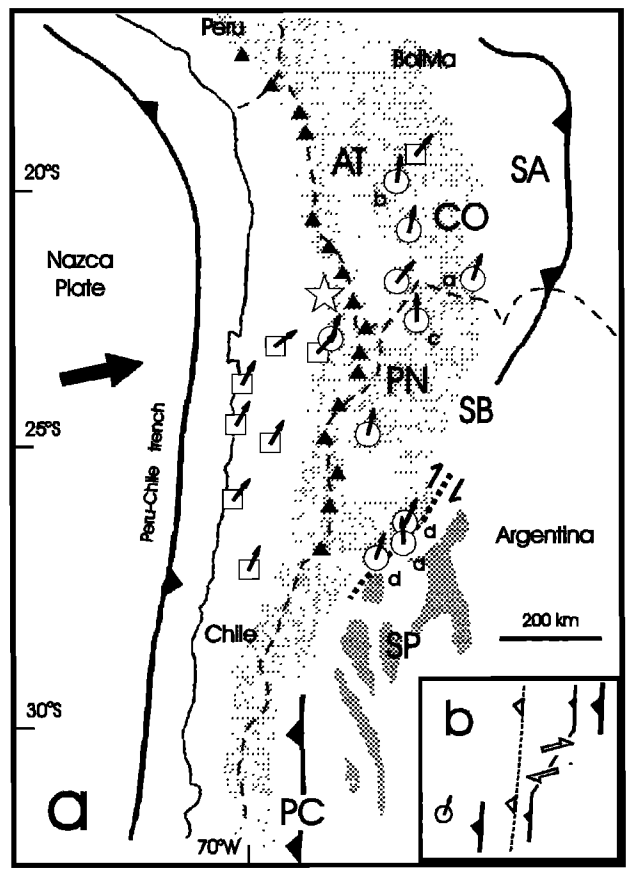


Figure 1. (a) Schematic map of the southern Central Andes showing the location of the Altiplano (AT) - Puna (PN) plateau. The eastern border of the plateau is marked by the Cordillera Oriental (CO). Light shading shows areas above 3000 m [after *Isacks*, 1988]. Small solid triangles denote the present volcanic arc [after *de Silva and Francis*, 1991]. Barbed lines indicate the Perú-Chile trench and the present orogenic front in the Subandean thrust belt (SA). Thick arrow indicates the mean direction of Nazca-South America plate convergence since the late Miocene [*Somoza*, 1998]. The thick-skinned foreland deformation belts are denoted by SP for the Sierras Pampeanas (dark shading showing basement uplifts) and SB for the Santa Bárbara System. PC is the Precordillera thin-skinned thrust belt of Argentina. Dashed line is the Tucumán dextral transfer zone. Symbols show localities with paleomagnetic information from Mesozoic-lower Tertiary (squares) and upper Tertiary (circles) rocks, with arrows representing the amount and sense of rotation (the north direction is taken arbitrarily as zero rotation). The star depicts the area where the upper Miocene rocks of this study were sampled. Upper Tertiary localities with labels denote data from upper middle Miocene or younger rocks: "a" is Quebrada Honda [*MacFadden et al.*, 1990], "b" is Quehua [*MacFadden et al.*, 1995], "c" is northwestern Puna [*Somoza et al.*, 1996], and "d" denote sites in the southern Puna [*Aubry et al.*, 1996]. Sources for the remaining paleomagnetic data from the southern Altiplano, Puna, and Cordillera Oriental are *Roperch et al.* [1993], *Butler et al.* [1995], *MacFadden et al.* [1995], and *Prezzi and Vilas* [1998]. Sources for the paleomagnetic data from northern Chile are *Hartley et al.* [1988, 1992], *Riley et al.* [1993], *Forsythe and Chishlom* [1994], *Dupont-Nivet et al.* [1996], and *Randall et al.* [1996]. (b) Scheme for the structural evolution in the Subandes of southernmost Bolivia during the last 10 m.y. following *Kley* [1996] and rotation of the Quebrada Honda section ("a" in Figure 1a). Dashed barbed line represents the initial orientation of early formed structures related to the Interandean thrust (time 1), thin solid barbed line represents the final orientation of the same structures (time 2). Eastward propagation of the Interandean thrust at time 1 is impeded in the southern area, while eastward propagation continues in the north on a newly formed lower thrust (thick solid barbed line in the upper right-hand corner of the diagram). Shortening in the south at time 2 occurs on an out-of-sequence thrust (thick solid barbed line in the lower left-hand corner of the diagram) located west of the Interandean structures. The north-south change in the location of shortening is accommodated by a dextral transfer zone in which earlier structures (time 1) are rotated clockwise (time 2). The affects of this dextral transfer zone extend toward the internal part of the orogen bending the western hanging wall block where the Quebrada Honda strata lie (circle shows relative location and amount of rotation). Figure is not to scale.

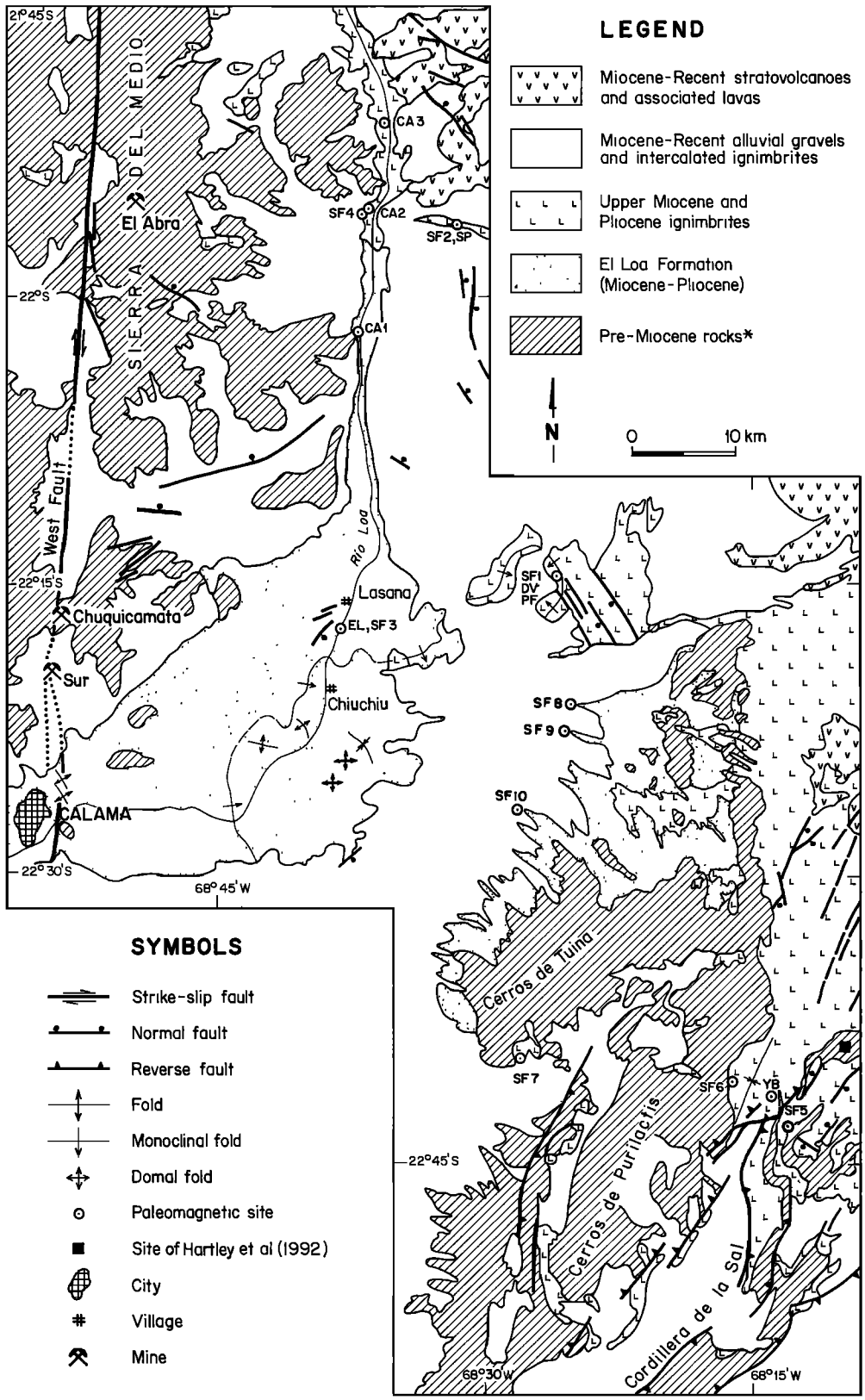


Figure 2. Simplified geologic map of the Calama Basin and Cerros de Tuina-Purilactis area, northern Chile, with only those structures affecting upper Miocene and younger units shown. Paleomagnetic sites are also located and labeled by unit and site number. EL, El Loa Formation; SP, Lower San Pedro Ignimbrite; DV, Divisoco Ignimbrite; PF, unnamed pyroclastic flow deposit below Divisoco Ignimbrite; SF, Sifón Ignimbrite; YB, Yerba Buena Ignimbrite; CA, Carcote Ignimbrite. Asterisk indicates that in the Cerros de Purilactis and Cordillera de la Sal this unit includes lower Miocene rocks. Simplified from A. Tomlinson et al. (unpublished mapping, 1999), *Marinovic and Lahsen* [1984], *Naranjo et al.* [1994], *Jolley et al.* [1990], and C. Arriagada (unpublished mapping, 1999).

rocks in this region. In order to attain to a better spatial-temporal resolution of the southern Central Andean paleomagnetic data set we performed a paleomagnetic study on upper Miocene rocks from the Precordillera of northern Chile (22°S). The age of the studied rocks is particularly pertinent because they are coeval with the initiation of the Subandean thrust belt and a major acceleration in uplift of the Central Andean plateau [*Benjamin et al.*, 1987; *Gubbels et al.*, 1993; *Kley et al.*, 1997], which *Isacks* [1988] relates to oroclinal bending.

The conclusion we reach is that our study area has not been affected by paleomagnetically determinable rotation since late Miocene time, suggesting that rigid-body rotation of entire northern Chile related to late Miocene-Recent shortening in the eastern foreland thrust belt is either of small magnitude or nonexistent. By inference, much of the rotation observed in upper Miocene and younger rocks in the southern Central Andes must be of local origin and related to local structures.

2. Regional Geology

The study area is located in the Miocene to Recent forearc region (Figure 1a). Pre-Miocene rocks in the area are principally composed of Upper Paleozoic to Eocene volcanic, granitic and sedimentary rocks [*Marinovic and Lahsen*, 1984; A. Tomlinson et al., unpublished data, 1999] which were deformed and uplifted during an important Eocene shortening event and transcurrent movement along the West Fault system (Figure 2) [*Reutter et al.*, 1996; *Tomlinson and Blanco*, 1997a]. Included in the basement rocks in Figure 2 are scarce exposures of Oligocene and lower Miocene continental-facies sedimentary rocks locally deformed by late Oligocene to early Miocene reverse faulting in the Cerros de Purilactis [*Hartley et al.*, 1992] and transcurrent faulting in the Sierra del Medio [*Tomlinson and Blanco*, 1997b].

The Sierra del Medio and Cerros de Tuina-Purilactis (Figure 2) limit the northwest and southeast sides, respec-

Table 1. Paleomagnetic Results

Site	Slat	Slong	n/n_o	α_{95}	k	D	I	Plat	Plong
			El Loa Formation at Lasana						
EL1	22° 17'	68° 38'	5/5	5.0	234	185.3	53.4	77.4	270.8
EL2	22° 17'	68° 38'	6/6	5.7	139	177.2	37.7	87.2	44.8
EL3	22° 17'	68° 38'	0/7	---	---	---	---	---	---
EL4	22° 17'	68° 38'	6/7	8.5	63	349.4	-41.9	80.1	8.2
EL5	22° 17'	68° 38'	7/7	8.4	52	187.6	53.4	76.5	263.1
EL6	22° 17'	68° 38'	4/4	9.3	58	357.9	-41.9	87.3	336.9
EL7	22° 17'	68° 38'	5/5	8.2	32	11.4	-47.0	78.1	233.7
			Sifón Ignimbrite						
SF1[a]	22° 14'	68° 26'	9/9	2.6	384	0.3	22.7	55.9	112.1
SF2	21° 56'	68° 32'	8/9	2.9	368	4.6	22.5	56.1	119.6
SF3	22° 17'	68° 38'	8/8	2.8	398	359.9	22.4	56.1	111.2
SF4	21° 56'	68° 37'	9/9	2.6	387	3.5	20.5	57.3	117.8
SF5[b]	22° 43'	68° 13'	8/9	4.7	140	358.0	27.5	52.7	118.6
SF6[c]	22° 40'	68° 17'	9/9	4.8	114	0.4	17.6	58.3	112.5
SF7[b]	22° 40'	68° 28'	5/7	3.7	328	8.1	19.9	56.1	125.9
SF8[d]	22° 21'	68° 25'	9/9	2.9	316	1.8	20.4	57.1	114.8
SF9[e]	22° 23'	68° 25'	7/7	3.3	295	358.0	18.5	58.7	116.3
SF10	22° 27'	68° 28'	7/8	4.4	187	2.5	17.2	58.1	107.8
Mean Sifón			10 sites	2.5	371	1.7	20.9	56.7	114.6
			Carcote Ignimbrite						
CA1	22° 02'	68° 37'	9/9	4.7	119	180.4	26.8	82.1	114.2
CA2	21° 56'	68° 37'	7/8	3.1	379	179.8	26.1	81.8	110.0
CA3	21° 51'	68° 36'	9/9	3.9	177	175.9	28.1	81.3	81.1
Mean Carcote			3 sites	3.6	1160	178.7	27.0	82.3	102.0
			Other Ignimbrites						
DV[a]	22° 14'	68° 26'	12/12	4.4	98	321.2	-40.5	54.3	12.5
SP	21° 56'	68° 32'	8/9	5.7	96	10.9	-26.1	76.8	165.0
PF[a]	22° 14'	68° 26'	4/4	9.4	118	15.4	-44.5	75.4	220.3
YB	22° 42'	68° 14'	5/5	5.7	180	162.9	75.6	48.2	303.4
Mean 1			n=12	13.8	11	359.2	-40.6		
Mean 2			n=11	10.0	22	358.9	-45.3		
Mean 3[f]			n= 6	7.2	88	0.9	-46.1		

Slat and Slong are site coordinates (south latitude, west longitude) to the nearest a minute, n/n_o is number of samples used in statistic/number of samples collected, α_{95} is semiangle of the 95% confidence cone about the site mean direction, k is the precision parameter [*Fisher*, 1953], D and I are declination and inclination referred to paleohorizontal, Plat and Plong are VGP coordinates (south latitude, east longitude), mean 2 is calculated without the Sifón Ignimbrite datum (as described in the text).

[a] After tectonic correction with strike 150° to the east and dip 11° (90° clockwise from strike) is applied.

[b] Site-mean obtained by combining demagnetization circles and stable end points.

[c] After tectonic correction with strike 18° to the east and dip 9° (90° clockwise from strike) is applied.

[d] After tectonic correction with strike 180° to the east and dip 4° (90° clockwise from strike) is applied.

[e] After tectonic correction with strike 240° to the east and dip 2° (90° clockwise from strike) is applied.

[f] El Loa Formation at Lasana

tively, of a broad, shallow Miocene-Pliocene basin, the Calama basin (Figure 2). The basin contains fine-grained playflat, palustrine carbonate, and lacustrine diatomite facies rocks at its center and coarser-grained fluvial facies material at the basin margins, all of which constitute the El Loa Formation [*Marinovic and Lahsen*, 1984; *May*, 1997]. Coarse alluvial gravels also occur along the basin margins and form piedmont deposits in the low areas within the surrounding ranges. Intercalated within these sedimentary layers, and locally forming extensive plateaus, are Miocene and Pliocene ignimbrites derived from volcanic centers near the international border or in southwestern Bolivia [*de Silva*, 1989].

3. Paleomagnetic Sampling

The goals of the paleomagnetic study were to identify the presence and magnitude of vertical-axis rotations in the area with respect to the craton and to determine the amount of relative rotation between sites within the study area. The latter can be accomplished by sampling ignimbrites with a large areal distribution [e.g., *Wells and Hillhouse*, 1989; *Hudson*, 1992]. Sample sites are shown in Figure 2, and their geographic coordinates are given in Table 1. Samples were collected using a portable field drill and were oriented using magnetic and solar compasses. Uncertainties for isotopic ages are quoted at 2σ, and those ages determined prior to 1977 are recalculated using the decay constants of *Steiger and Jäger* [1977]. Weighted mean ages are weighted by the inverse of the analytical variance, and its uncertainty is 2σ of the population [*McDougall and Feibel*, 1999], which we consider is a more realistic estimate of the age uncertainty than the standard deviation of the mean. The timescale used is that of *Berggren et al.* [1995].

The El Loa Formation was sampled in a 45-m-thick section that crops out south of Lasana in the Rio Loa canyon (sites EL in Figure 2), where the base of the unit is not exposed and the top is overlain by the Sifón Ignimbrite. The dominant rock types in this section are conglomerates and coarse-grained sandstones, with decimeter-scale fine-grained beds being subordinate. Forty-one paleomagnetic samples were taken from seven beds (sites) of tan fine-grained sandtones, mudstones, and calcareous mudstones. Approximately a kilometer south of the EL sites, in the same stratigraphic section, *May* [1997] obtained a weighted mean $^{40}Ar/^{39}Ar$ age of 11.32 ± 0.28 Ma (2σ of the population) for single-crystal laser fusion of five biotite grains from a fallout tuff located 17 m stratigraphically below the Sifón Ignimbrite. This places the upper part of the sedimentary section near the middle-upper Miocene boundary.

Of the ignimbrites sampled, the oldest are the Lower San Pedro and Divisoco ignimbrites and an unnamed ignimbrite underlying the Divisoco. K/Ar dating indicate that these units were deposited between 11.3 and 10.3 Ma [*Baker*, 1977a, b; *de Silva*, 1989; A. Grunder and A. Tomlinson, unpublished data, 1999]. The full distribution of these older ignimbrites is unknown as they crop out only in canyons where Pliocene-Quaternary drainages have dissected the upper Miocene gravel-ignimbrite succession, but their distribution appears to be limited, as they are recognized only near their type sections [*Baker*, 1977a; *de Silva*, 1987, 1989]. Eight samples were taken from the Lower San Pedro Ignimbrite (site SP in Figure 2), and 12 samples were taken from the Divisoco Ignimbrite (site DV in Figure 2) within and near their type sections. The ignimbrite underlying the Divisoco Ignimbrite has not been previously described (site PF in Figure 2). It is a gray, nonwelded, pumaceous pyroclastic flow deposit with a thickness of 14 m and block-size pumice up to 30 cm in diameter. Paleomagnetic samples were taken from four pumice blocks.

The most extensive ignimbrite in the study area is the Sifón Ignimbrite [*Guest*, 1969], which was collected at 10 sites (SF sites in Figure 2). The sites are distributed over much of the ignimbrite's known area of exposure [*de Silva and Francis*, 1989], covering ~90 km latitudinally. Sites SF1, SF2, and SF3 were taken from sections where the unit conformably overlies sites DV-PF, SP, and EL, respectively, and site SF4 is located in a section where the ignimbrite conformably underlies site CA2. A weighted mean of 24 published K/Ar ages (4 hornblende and 20 biotite) [*Baker*, 1977a; *Marinovic and Lahsen*, 1984; *de Silva*, 1989; *Naranjo et al.*, 1994] from this unit gives 8.3 ± 0.4 Ma, which is concordant with the $^{40}Ar/^{39}Ar$ biotite age of 8.27 ± 0.13 Ma obtained by *May* [1997].

The Yerba Buena Ignimbrite was sampled at one site (site YB in Figure 2), where it conformably overlies the Sifón Ignimbrite. *de Silva* [1989] reports biotite K/Ar dates of 8.2 ± 0.2 Ma and 8.5 ± 0.6 Ma from this unit.

The youngest ignimbrite sampled is the Carcote Ignimbrite [*Baker*, 1977a]. In the northern part of the study area, and further north, this ignimbrite is widely distributed [*Baker*, 1977a]. It was collected at three sites constituting 26 samples. A weighted mean of eight K/Ar biotite ages [*Baker*, 1977a; *Christensen et al.*, 1969; A. Tomlinson et al., unpublished data, 1999] from this unit is 5.8 ± 0.4 Ma. We will show later in this paper that the characteristic magnetization of this ignimbrite is one of reversed polarity (see Figure 6a). This additional constraint indicates that its deposition occurred during chron C3r (5.9-5.2 Ma).

4. Structural Geology

Many of the structural systems in Figure 2 initially became established during important Eocene-early Oligocene and late Oligocene-early Miocene deformations, when much of the current basin-and-range-style topography formed. In contrast, subsequent late Miocene and younger deformation is relatively minor; however, there are three areas where it is concentrated and more important: the West Fault system, the Cerros de Purilactis, and the central area of the Calama basin, near Chiuchiu.

The West Fault system had important strike-slip displacement events (kilometers to tens of kilometers) in the Eocene-early Oligocene and late Oligocene-early Miocene [*Reutter et al.*, 1996; *Tomlinson and Blanco*, 1997a, b], but crosscutting relations with younger units indicate it was reactivated in the late Miocene and Pliocene, when it produced offsets of >100 m dip-slip and ~500 m dextral strike-slip in Miocene gravels [*Tomlinson and Blanco*, 1997b; A. Tomlinson, unpublished data, 1999]. The scarcity of datable intercalated ashes and the nonfossilliferous nature of the strata make it difficult to constrain the full timing of the fault activity; however, dextral strike-slip, normal, and reverse faults cutting middle to upper Miocene gravels in the Sierra del Medio both predate and postdate the 5.8 Ma Carcote Ignimbrite, indicating that the various movements are broadly contemporaneous and late Miocene in age [*Tomlinson and Blanco*, 1997b]. En echelon

folds affecting the highest levels of the El Loa Formation near Calama (Figure 2) suggest that dextral shear continued into the Pliocene.

In the Cerros de Purilactis the most significant episodes of deformation occurred in the Eocene and latest Oligocene-early Miocene and involved reverse faulting, folding, and uplift [*Hartley et al.*, 1992]. Structural relations with younger units indicate that the reverse faults bounding the range on its southeast and northwest margins (Figure 2) were reactivated in the late Miocene-early Pliocene, and again in the late Pliocene-Quaternary on the southeast margin [*Marinovic and Lahsen*, 1984; *Jolley et al.*, 1990; *Naranjo et al.*, 1994], with this latter phase also affecting the Cordillera de la Sal [*Jolley et al.*, 1990; *Wilkes and Görler*, 1994]. At the SF6 site the first reactivation event folded the Sifón and Yerba Buena Ignimbrites [*Marinovic and Lahsen*, 1984]. *Lahsen* [1982] reports this pulse of shortening postdates ignimbrites as young as 7 Ma but predates the unsampled 4.2 Ma Puripicar Ignimbrite.

In contrast to the above structures, folds within the Calama basin have a geometry indicating that they are not of regional tectonic origin [*Naranjo and Paskoff*, 1981]. The area of folding is bound by a monocline having an irregular semicircular trend (Figure 2) which affects the El Loa Formation in the Chiuchiu area and the Miocene ignimbrite-gravel succession in the area of sites SF1-DV-PF. Outside of the monocline-bound area the El Loa Formation is subhorizontal and largely undeformed, but inside the area the unit is complexly folded into domes, depressions, and doubly plunging folds with irregular axial trends. The similarity of the structures to those found in salt fields and occurrence of evaporites within the unit suggest a salt tectonic origin for the structures [*Naranjo and Paskoff*, 1981]. These structures predate deposition of upper Pliocene-Pleistocene lacustrine rocks [*Naranjo and Paskoff*, 1981].

5. Paleomagnetism: Methods and Results

5.1. Demagnetization of the Natural Remanent Magnetization (NRM)

One to three specimens of each sample were subjected to stepwise demagnetization. Most magnetization measurements were done with a three-axis 2G Enterprises cryogenic magnetometer. Alternating field (AF) demagnetization employed either a 2G600 or a prototype tumbling AF demagnetizers. Thermal demagnetization used a Schonstedt TSD-1 furnace. Bulk magnetic susceptibility was measured after each heating step in order to monitor possible magnetic mineral transformations. Demagnetization results were plotted on orthogonal demagnetization diagrams and equal-angle projections and were analyzed by both visual inspection and vector subtraction. Magnetic components and their quality statistics were determined using principal components analysis [*Kirschvink*, 1980; *Torsvik*, 1992]. Specimens showing a maximum angular deviation (MAD) >10° were rejected from further study. Site-mean directions were calculated using standard Fisherian statistics [*Fisher*, 1953]. Results for two sites, however, were

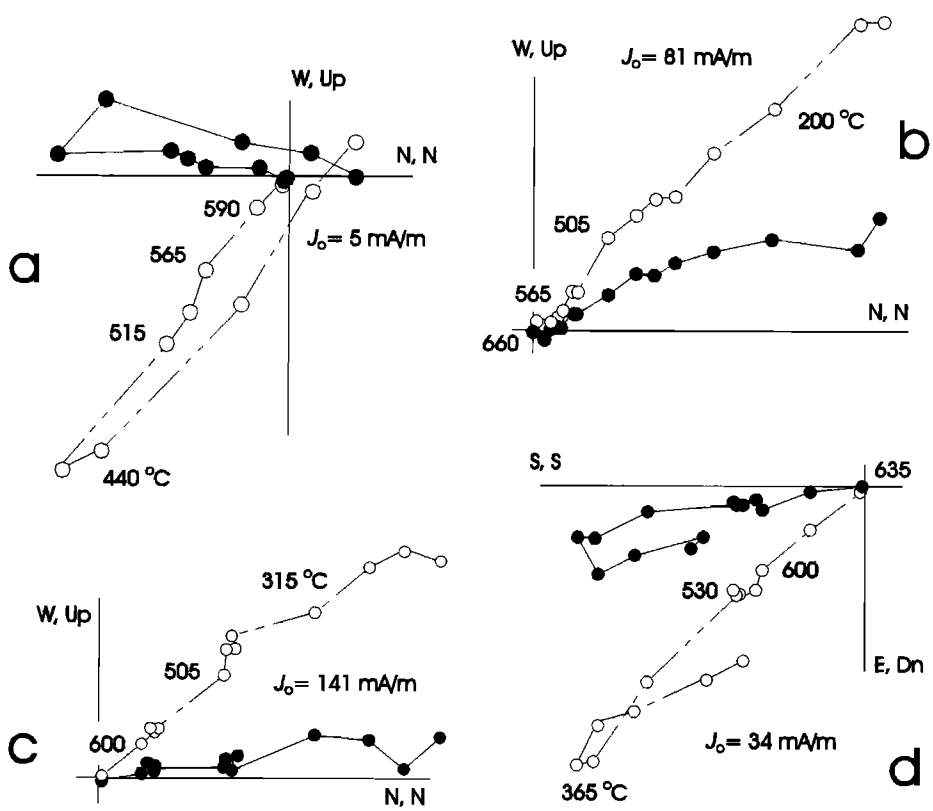


Figure 3. Examples of typical thermal demagnetization behavior from rocks of the El Loa Formation at Lasana. Samples are from the sites (a) EL5, (b) EL4, (c) EL6, and (d) EL2. Open (solid) circles indicate vertical (horizontal) projection.

determined by combining demagnetization circles and stable end points [*McFadden and McElhinny*, 1988]. Paleomagnetic results are given in Table 1.

5.1.1. Sedimentary rocks (Figure 3). Specimens of the El Loa Formation were thermally demagnetized. AF demagnetization up to 120 mT was performed in a few cases, but generally it was followed by additional thermal steps. One site did not give reliable results and was rejected from further analysis. At three sites, a southward, downward component (reversed polarity) was observed after elimination of a softer vector usually between 200° and 400°C (e.g., Figures 3a and 3d). This soft component is close to the present dipolar field in the region and in site EL5 shows greater magnitude than the reverse component isolated at higher temperatures (Figure 3a). Magnetization in the remaining three sites is entirely of normal polarity (e.g., Figures 3b and 3c). In this case it is difficult to distinguish the temperature at which any soft, present-field component was removed. This is because the rocks are subhorizontal and the late Miocene geomagnetic field for South America is almost indistinguishable from the present dipolar field. Since rock magnetic results and microscopic observations indicate similar behavior and mineralogy between different sites of the Lasana section (see section 5.2), we adopted the temperature that eliminates most of the soft component in site EL5 (near 400°C, Figure 3a) to start the definition of the component of interest for the normally magnetized sites.

Paleomagnetic results for the El Loa Formation at Lasana

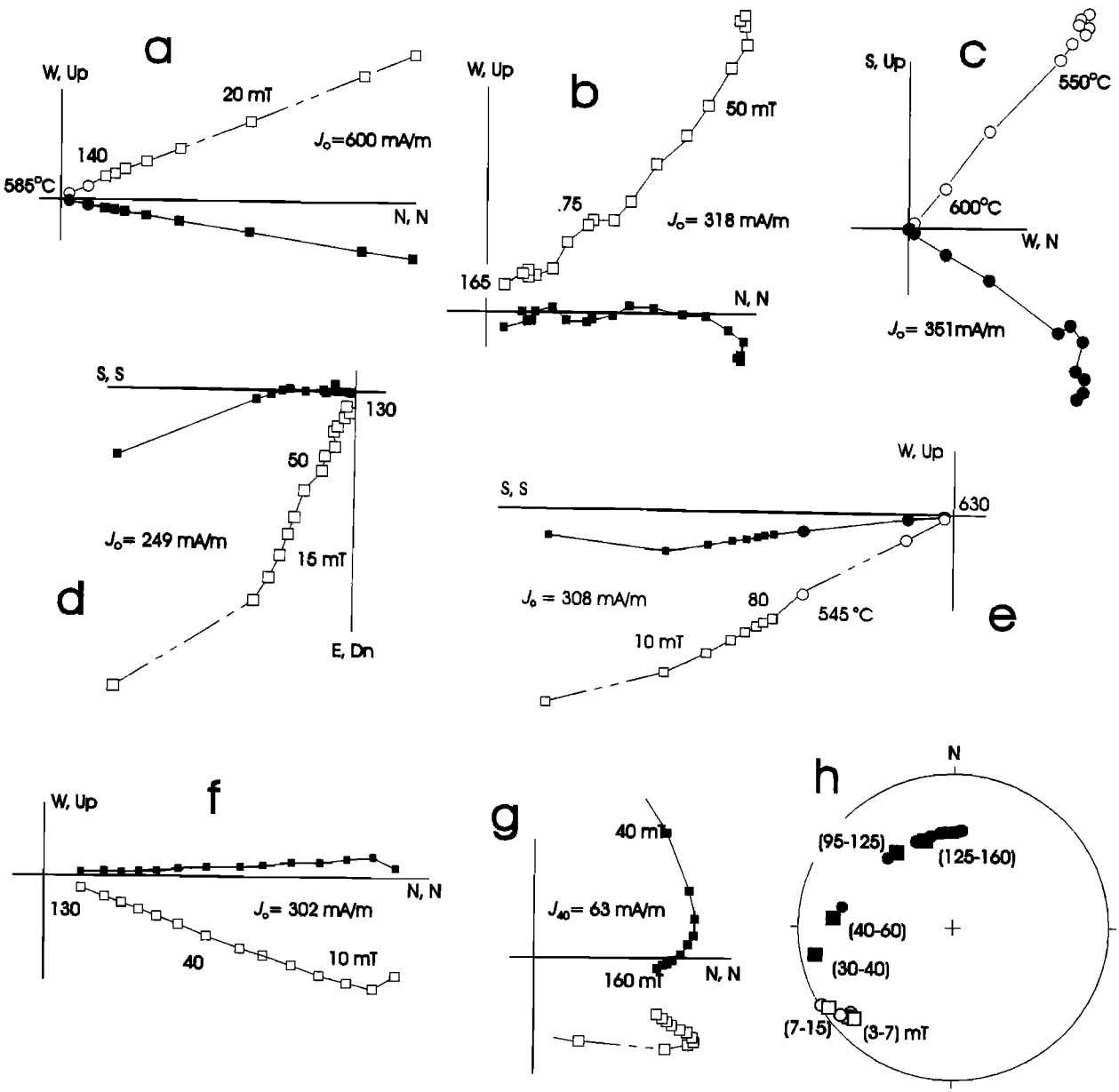


Figure 4. Ignimbrites: Figures 4a to 4f are examples of single-component paleomagnetic behavior of samples from the sites (a) SP, (b) PF, (c) DV, (d) YB, (e) CA2, and (f) SF3. Figures 4g and 4h show multicomponent behavior at site SF5: (g) partial orthogonal plot (from 40 to 160 mT) showing the persistence of coercivity spectra overlap at high AF steps; and (h) stereoplot showing the path of partial remanence (circles) and the path of vector subtraction between successive steps (squares) for the same specimen. Convention in orthogonal plots as in Figure 3, with squares (circles) indicating AF (thermal) steps.

(Table 1) indicate that this short section contains at least four magnetic polarity zones (from bottom to top of the section: R-N-R-N).

5.1.2. Ignimbrites (Figure 4). Both AF and thermal demagnetization were successful in isolating a stable component of magnetization in specimens from most sites of the ignimbrites. These components form linear segments on orthogonal plots which decay toward the origin (Figure 4). Complete demagnetization of specimens from the site CA2 was possible only after thermal cleaning (Figure 4e).

Thermal demagnetization (up to 630°C) in samples from the sites SF5 and SF7 determined a poorly clustered, up-to-the-south component, which is also observed in the earliest steps of static AF demagnetization. The presence of the Sifón Ignimbrite characteristic magnetization was detected at higher AF steps, although coercivity spectra overlap (Figure 4g) and acquisition of gyroremanent magnetization hampered its definition. Tumbling AF demagnetization, however, allowed the incipient isolation of the harder vector in some specimens, which was defined after subsequent thermal treatment. These vectors were combined with demagnetization circles (e.g. Figure 4h) to compute the site-mean magnetization for the SF5 and SF7 sites.

5.2. Optical Observations and Rock Magnetic Properties

Petrographic and rock magnetic analyses were conducted on selected samples to identify the principal carriers of remanence and their magnetic properties. Observations of polished sections were made through oil immersion in reflected light. Oxide grains range from 260 μm in the ignimbrites, and 100 μm in the sedimentary rocks, down to the optical resolution of the microscope. Most of these grains are titanomagnetites showing a variety of high-temperature (above 600°C) oxidation textures, which were classified following *Haggerty* [1991].

Titanomagnetites in the ignimbrites occur as both inclusions in mafic silicates and free in the glass matrix. The main oxidation products are fine to coarse ilmeno-hematite lamellae intergrowths along the {1 1 1} planes of spinel. The lamellae occur either homogeneously distributed in the phenocrysts (e.g., SF2, SF6, and CA1 sites) or preferentially concentrated along cracks and grain boundaries (e.g., DV and SP sites). Typical oxidation textures in these sites correspond with the C2, C3, and, to a lesser degree, the C4 stages of *Haggerty* [1991]. Higher oxidation products are present in the SF5 site (C4 stage strongly dominant) and CA2 site (C5-C6 stages).

Titanomagnetites in the sedimentary rocks occur as moderately rounded grains which are almost identical to, and represent the full spectrum of, oxide grain varieties observed in the ignimbrites.

Bulk rock hysteresis curves and isothermal remanent magnetization acquisition (IRM) were performed using a Molspin vibrating sample magnetometer. Saturation magnetization (J_S), saturation remanence (J_R) and coercivity (H_C) were determined from the hysteresis loops (induction up to 1 T), whereas coercivity of remanence (H_{CR}) was obtained after back field of IRM acquired at 0.7 T. Strong field thermomagnetic analyses were carried out in air with a horizontal Curie

Table 2. Rock Magnetism Results

Site	J_o	χ	H_C	H_{CR}	H_{CR}/H_C	J_S	J_R	J_R/J_S	T_C	Three Orthogonal Axes IRM		
	mA/m	10^{-3} SI	mT	mT		10^{-2}Am^2/Kg	10^{-2}Am^2/Kg		°C	L, % (°C)	M, % (°C)	H, % (°C)
EL1	40	4.1										
EL2[a]	110	3.8	11.0	33.0	3.0	10.4	1.60	0.16	563	67 (580)	22 (580)	11 (640)
EL3	130	9.0										
EL4[a]	70	5.6	9.5	27.0	2.8	7.6	0.85	0.11		68 (560)	24 (600)	8 (630)
EL5[a]	7	3.2	6.0	25.0	4.1	18.2	1.60	0.09	534	68 (600)	24 (600)	8 (630)
EL6	110	8.0	15.0	38.0	2.5	20.4	3.80	0.18	534	74 (580)	14 (600)	12 (630)
EL7	100	8.7										
SF1	280	7.0	12.5	40.0	3.2	64.6	5.30	0.08	565			
SF2[a]	260	8.0								65 (540)	31 (580)	4 (>600)
SF3	320	5.0										
SF4	170	6.0										
SF5[a]	3600	2.3	16.0	42.5	2.6	25.2	2.90	0.11	567	45 (580)	25 (580)	30 (610)[b]
SF6[a]	145	3.5	18.0	46.0	2.5	50.2	6.00	0.12				
SF7	8600	4.3	16.0	41.0	2.6	75.5	10.00	0.13		50 (550)	47 (550)	11 (630)
SF8	288	4.0										
SF9	143	4.9										
SF10	227	5.9										
CA1[a]	90	1.4	5.0	28.0	5.6	7.4	0.30	0,04	541	67 (520)	21 (560)	12 (630)
CA2[a]	320	0.2	14.5	63.0	4.3	2.1	0.30	0.15		31 (520)	22 (600)	47 (630)
CA3	780	1.7										
PF	300	3.4	2.0	31.0	15.5	37.9	0.60	0.02	510			
YB	220	6.4	6.0	33.0	5.5	59.0	3.60	0.06				
SP[a]	510	1.9	11.5	36.0	3.1	30.3	3.00	0.10	550	71 (560)	26 (580)	3 (>600)
DV[a]	80	3.7	7.0	32.0	4.6	28.5	1.60	0.06	565	72 (500)	17 (540)	11 (630)

J_0 is NRM, χ is arithmetic mean susceptibility, H_C is coercivity, H_{CR} is coercivity of remanence (after back-field of IRM acquired at 0.7 mT), J_S is magnetization at 1 T, J_R is magnetization remanence, T_C is Curie temperature. Results of thermal demagnetization of three orthogonal axes IRM, with L, M, and H being the low (< 0.11 T), medium (0.11-0.45 T) and high (0.45-2.3 T) coercivity fractions, respectively. The values indicate the initial percentage of magnetization acquired on each axis respect to the arithmetic sum of the three axes, and the approximate temperature after which the remanence on each axis is only 15-10% of the initial value.

[a] Iron-titanium mineralogy petrographic observations.

[b] The high coercivity fraction shows an early drop to 30% of initial remanence at 320°C.

balance, where heating (up to 700°C) and cooling rates were of the order of 80°C/min. Curie temperatures (T_C) were determined by the method of *Grommé et al.* [1969]. Stepwise thermal demagnetization of IRM on three orthogonal axes [*Lowrie*, 1990] was used in order to identify high (0.45-2.3 T), intermediate (0.11-0.45 T), and low (< 0.11 T) coercivity minerals. Table 2 lists bulk rock magnetic characteristics, and Figure 5 illustrates some representative behaviors.

The rock magnetism results indicate the dominance of low-coercivity minerals with the subordinate presence of a high-coercivity phase in most studied samples. The contribution of high-coercivity minerals is important in the SF5 site, and it dominates the remanence in the CA2 site. Strong field thermomagnetic experiments show near reversible curves (e.g., Figure 5b). The Curie temperatures of the analyzed samples (Table 2) suggest that Ti-poor magnetite is the dominant ferrimagnetic mineral.

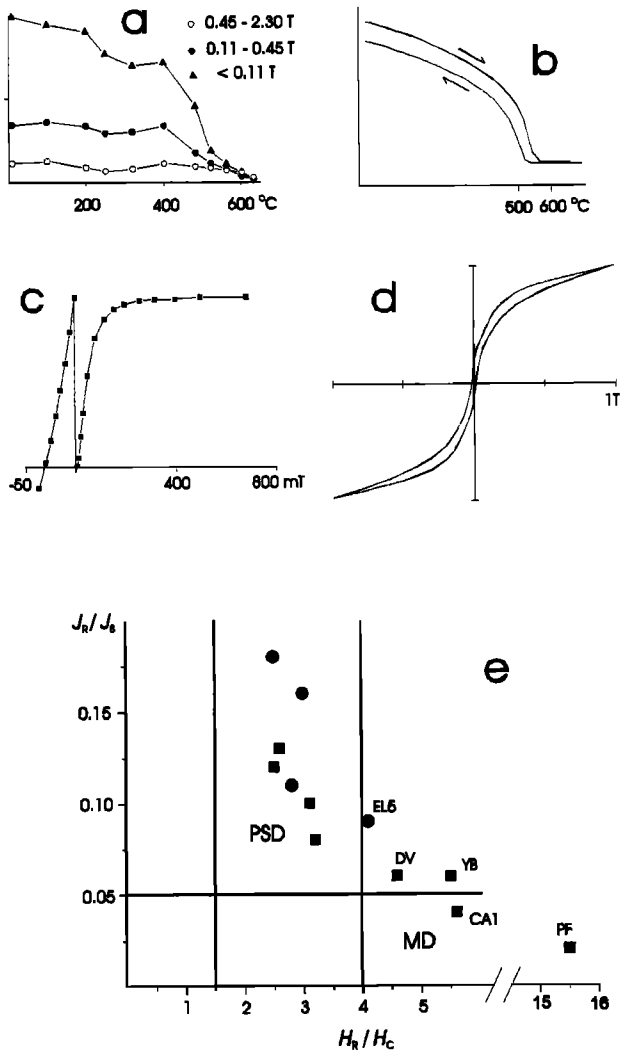


Figure 5. Representative rock magnetic behaviors. (a) The low- and medium-coercivity fractions are the main carriers of the IRM at the site EL4; (b) nearly reversible strong field thermomagnetic curve from a sample of the site SP; (c) IRM acquisition and back field curves from a sample of the site DV (note the change of scale in the abscissa); (d) wasp-waisted shape of the hysteresis loop due to a mixture of ferrimagnetic and antiferrimagnetic minerals at the site CA2; and (e) *Day et al.* [1977] classification of domain state for titanomagnetites with circles (squares) showing data from sedimentary rocks (ignimbrites). The sites plotting out of the PSD region are identified (see text for discussion). The sites CA2 and SF5 have an important quantity of hematite-like minerals, and so they are not included in this diagram.

5.3. Origin of the Magnetizations

The results of the paleomagnetic, rock magnetism and petrographic studies argue for a primary origin of the magnetizations in Table 1. Petrographic observations indicate that the magnetic mineralogy of these rocks is dominated by high-temperature products of oxidation-exsolution in primary titanomagnetite, such as Ti-poor titanomagnetite and titanohematite. No evidence of low-temperature oxidation products (e.g., titanomaghemite) were found. The observed titanomagnetite grain size ranges from 260 μm down to the optical resolution of the microscope, the smaller size fraction probably having pseudo-single domain (PSD) characteristics. Additionally, the trellis ilmenite-lamellae isolate small areas of titanomagnetite poorer in Ti within the larger grains, diminishing the effective magnetic grain size and possibly increasing the magnetization's stability [e.g., *Strangway et al.*, 1968; *Ade-Hall*, 1969].

The magnetic signal of the hysteresis loops must reflect the sum of the weighted contributions of minerals of different grain size and composition [e.g., *Roberts et al.*, 1995]. Figure 5e shows that seven out of 12 observed titanomagnetite-dominant sites have hysteresis parameters within the PSD area of the hysteresis-related domain-state classification of *Day et al.* [1977]. Those sites plotting out of the PSD region must have a higher content of multidomain (MD) grains.

The paleomagnetic characteristics of the studied rocks is the most convincing argument for the presence of stable magnetizations. Magnetizations from specimens of all sites but one (EL3) defined a well-clustered group of vectors within each site. The in-situ magnetizations observed in 19 out of 23 sites are different from the present dipolar field at the study area (Figure 6c), suggesting they are ancient magnetizations. Magnetization at sites PF (in-situ), EL4, EL6, and EL7 are indistinguishable from the present dipolar field (Figure 6 and Table 1). The latter three, however, have PSD-like rock magnetic parameters and paleomagnetic behavior similar to the reversely magnetized sedimentary sites, suggesting that they are also ancient magnetizations. Likewise, AF demagnetization of samples from the PF site (Figure 4b) indicates that any soft contribution of the larger MD grains to the NRM is negligible.

We conclude that the magnetizations listed in Table 1 for the ignimbrites are of thermoremanent origin, linked to emplacement and cooling of each flow.

Deposition of the El Loa Formation occurred during times of important volcanic activity in the region, as is indicated by the occurrence of ignimbrites and fallout ash tuffs intercalated within the unit [*Marinovic and Lahsen*, 1984; *May*, 1997]. Both optical observations and rock magnetism measurements indicate the magnetic mineral assemblages are similar in the ignimbrites and sedimentary rocks studied. This and the volcanic provenance character of the sedimentary rocks, as indicated by the presence of volcanic lithic grains, sanidine, and ferromagnesian minerals, suggest that the magnetic grains of these rocks are derived from erosion and reworking of vol-

canic deposits in the area. Furthermore, the degree of oxidation in opaque grains from the El Loa Formation is similar to that observed in the ignimbrites, suggesting that oxidation predates deposition. Thus a detrital origin for the magnetizations isolated in the El Loa Formation at Lasana is suggested.

6. Tectonic Significance of the Results

6.1. Relative Rotations Between Sites and Structural Integrity of the Study Area

Paleomagnetic data from individual sites of the Carcote and Sifón ignimbrites were compared to assess relative rotations between sites and, by inference, to assess the structural integrity of the studied area. Magnetizations from the Carcote Ignimbrite document no significant relative vertical-axis rotations among the three sites studied (Figure 6a); consequently, a mean direction was obtained by averaging magnetizations from these sites (Table 1).

Paleomagnetic directions from 10 sites of the Sifón Ignimbrite are well clustered (Figure 6b), with the dispersion in the declinations being similar to the dispersion in the inclinations. However, directions from sites SF5 and SF7 are in a marginal position around a clustered central group. The somewhat greater dispersion of the directions from these sites could be related with the difficulties in determining their characteristic magnetizations. An alternative explanation for the site SF5 is the recording of a slightly different geomagnetic field. This site is in the thicker of the sampled sections of the Sifón Ignimbrite, where thermal effects, such as a densely welded vitrophyre, are the most evident [*Guest*, 1969]. Accordingly, we consider the possibility of a later acquisition of the thermoremanent magnetization due to slower cooling.

The remaining eight directions of the Sifón Ignimbrite, although well clustered, reveal minor discrepancies when they are compared one to another. Geologic observations suggest that this ignimbrite is a simple cooling unit [*de Silva*, 1989], and we suggest that it carries a primary thermoremanent magnetization at the studied sites. If so, some initial between-site dispersion of the thermoremanent magnetization may have been produced by subblocking-temperature plastic deformation [e.g., *Geissman*, 1980; *Rosenbaum*, 1986] and/or intraunit magnetic field variations at the time of cooling [e.g., *Doell and Cox*, 1963; *Baag et al.*, 1995]. The possibility of these factors producing small intraunit dispersion in thermoremanent magnetizations is suggested by paleomagnetic results from historic volcanic units [e.g., *Hoblitt et al.*, 1985; *Castro and Brown*, 1987]. Further between-site dispersion of the paleomagnetic data can result from small-block vertical-axis rotations, which is the effect we want to quantify, and/or undetected or improper structural correction at the site. The clustering of the central group in Figure 6b indicates that the dispersion produced by any combination of these factors is small, and so unambiguous vertical-axis rotations between sites are not detected. Furthermore, the angular standard deviation of the site-mean directions in Figure 6b (δ = 4.2°, n=10) is comparable with that obtained after widespread sampling of undeformed ignimbrites elsewhere [e.g., *Rosenbaum*, 1986; *Wells and Hillhouse*, 1989; *Hudson*, 1992; *Palmer et al.*, 1996]. Thus we interpret the results from the Sifón Ignimbrite as indicating that any occurrence of vertical-axis rotation affecting a particular site(s) is minor or nonexistent. A mean-group direction (n=10) for the unit is given in Table 1. This overall mean is similar to that obtained after excluding the SF5 and SF7 directions (D=1.4°, I=20.2°, α_{95}=2.1°, k=726, δ=3°, n=8).

The SF sites are irregularly distributed over a N-S strip of 90 km (Figure 2). The paleomagnetic results suggest that no relative vertical-axis rotations have occurred among the 10 sites since 8.3 Ma, the age of the Sifón Ignimbrite. Furthermore, since rocks of the EL, PF, SP, DV, and CA2 sites are from localities where the Sifón Ignimbrite was sampled, they too have not experienced relative vertical-axis rotations since 8.3 Ma. Relative rotations between sites prior to deposition of the Sifón Ignimbrite can not be excluded, but any rotation would have had to occur without significant tilting as each of the sections are conformable.

The lack of relative vertical-axis rotations between sites means either that the entire study area has acted as a single coherent structural block with respect to vertical-axis rotational deformation or the deformation is homogeneous. The latter is rejected as it implies a homogeneous structural style, which is not the case for the late Miocene and younger structural geology in the area (Figure 2). Therefore we conclude that with respect to vertical-axis rotation the area has acted as a single coherent structural block since the late Miocene, having dimensions of at least 90 km latitudinally and 40 km longitudinally. Any late Miocene-Recent rotation with respect to the craton would have to have affected the entire area.

6.2. Averaging Paleosecular Variation (PSV) and Comparison With Reference Poles

Our data set comprises 12 independent records of the paleofield (Figure 6c). The computed mean direction (mean 1, n=12) is given in Table 1. The directions of magnetization of the six ignimbrites are more scattered than those from the six sedimentary sites, reflecting their different mechanisms in acquiring the remanence. The ignimbrites provide spot recordings of the paleomagnetic field at their time of deposition, thus having a greater chance to record large PSV. This is the case for several units in this study. In particular, the characteristic magnetization of the Sifón Ignimbrite is strongly removed (62°) from the overall mean direction. Consequently, a more suitable mean direction for tectonic analysis (mean 2 in Table 1) was computed after excluding the Sifón Ignimbrite datum.

In the case of the sedimentary rocks the acquisition of remanence is slower, and so the PSV can be partially averaged within the site. Furthermore, the paleomagnetic results from the El Loa Formation at Lasana show the presence of four polarity zones and yield a positive (class "C") reversal test [*McFadden and McElhinny*,1990], suggesting that PSV sampling was acceptably attained in this section. We consider that the mean direction of this sedimentary section (mean 3 in Table 1) gives the best estimate of the late Miocene geomagnetic field in the region. In any case, mean 2 and mean 3 are separated by only 1.6°. The two pole positions derived from both these subgroups of data were compared with coeval paleopoles of stable South America.

Although agreement exists that the late Miocene reference pole for South America is close to the present geographic pole, previous Andean paleomagnetic studies have used slightly different paleopoles. We compare our results with two previously proposed, slightly different, late Miocene paleomagnetic poles (paleopole A after *Somoza et al.* [1996]; paleopole B after *Aubry et al.* [1996]) and with a paleopole coincident

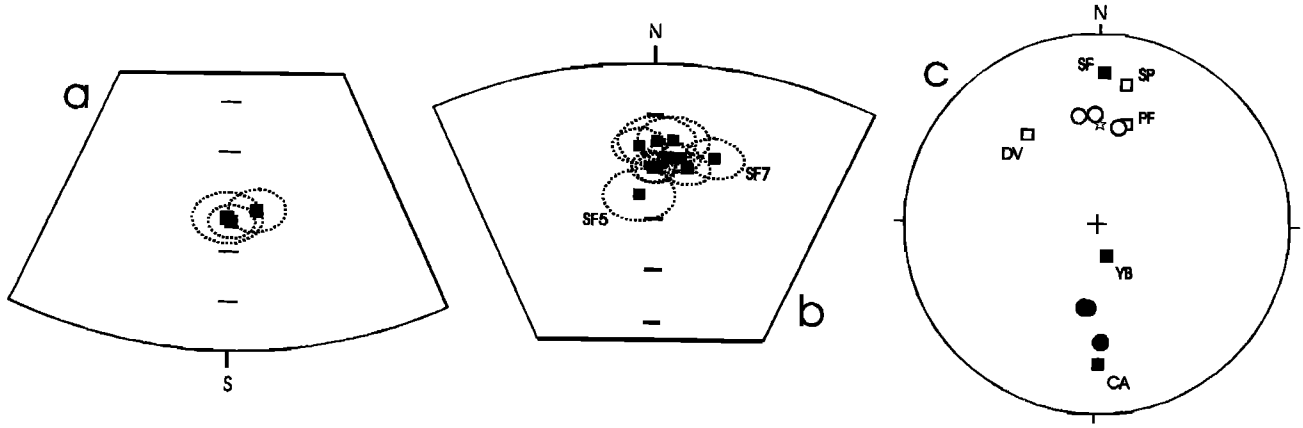


Figure 6. Equal-area projection of the site-mean magnetizations. Solid (open) symbols indicate lower (upper) hemisphere projection. Dotted lines in Figures 6a and 6b denote the α_{95} cones of confidence around the individual site-mean directions. (a) Site-mean directions from the three Carcote Ignimbrite sites; (b) site-mean directions from the ten Sifón Ignimbrite sites, directions in the outer fringe of the central cluster are identified (see text for discussion); and (c) the 12 independent records of the paleofield determined in this study. Circles (squares) represent sedimentary rocks (ignimbrites). Star denotes the present dipolar geomagnetic field. Labels identify the ignimbrites.

with the present geographic pole (paleopole C) using an arbitrary A_{95} of 3°. We prefer using either the A or B reference poles, rather than the paleopole C, since they are derived from paleomagnetic data. Rotation (R), paleolatitude anomaly (P), and their 95% confidence limits were calculated in VGP space by the method of *Beck et al.* [1986]. The results are given in Table 3. The somewhat large confidence limits of our data sets and uncertainty as to which reference pole is the correct one to use preclude a more precise definition of rotation. Nevertheless, the calculated rotation values indicate that regardless of which data set or reference pole used, no paleomagnetically determinable rotation with respect to the craton has affected the region since the late Miocene.

Sites SF5 and SF6 are located 10 km southwest of lower Miocene rocks for which about 20° clockwise rotation were reported by *Hartley et al.* [1992] (Figure 2). They suggested that most (if not all) of this rotation took place during the early Miocene, which is in agreement with our results.

7. Paleomagnetically Determined Rotations From Upper Miocene Rocks in the Southern Central Andes: Local Accommodations of Regional Deformation

The aim of this section is a tectonic analysis of late Miocene-Recent rotations in the southern Central Andes. This time span is selected because it comprises the history of the rocks affected by this study and major tectonic events in the region, such as the initiation of the Subandean thrust belt and a major acceleration in uplift of the Central Andean plateau [e.g., *Benjamin et al.*, 1987; *Gubbels et al.*, 1993; *Kley et al.*, 1997], which in turn were related to oroclinal bending by *Isacks* [1988]. In order to rule out any possible older than late Miocene rotation, only paleomagnetic data from upper middle Miocene (circa 12 Ma) and younger rocks are discussed (those labeled in Figure 1a). The rotation amounts cited below are with respect to both reference pole A and reference pole B of Table 3, reference pole C yielding values 2° less.

Late Cenozoic horizontal shortening in the southern Central Andes shows variations in tectonic style, spatial distribution, and timing [e.g., *Allmendinger and Gubbels*, 1996], with the amount of shortening decreasing from Bolivia toward the south. *Isacks* [1988] presented a forearc oroclinal model assuming that the differential shortening varies smoothly along the orogen and accommodated a rigid-body, clockwise rotation of northern Chile. Isacks envisaged that homogeneous, large-scale rotation of the forearc (i.e., without important internal deformation) is possible due to its long and slender shape. Our results could be reconciled with Isacks' model 1 (his preferred model), which invokes only 5° of clockwise rotation for the last 12 m.y. Updating the estimate of differential shortening using recently published structural data suggests that any rigid-body-like rotation of northern Chile may be even less for this time span [*Somoza et al.*, 1997]. These low

Table 3. Paleopoles, Rotation and Paleolatitude Anomalies

	Lat	Long	A_{95}	K	N	Reference Pole A		Reference Pole B		Reference Pole C	
	°S	°E	deg.			R	P	R	P	R	P
Pole 2	84	301	10.9	19	11	1.1 ± 11.0	-3.7 ± 13.0	1.4 ± 10.0	0.4 ± 12.0	-1.0 ± 10.0	-6.1 ± 12.0
Pole 3	84	280	7.5	82	6	3.0 ± 8.0	-4.5 ± 6.2	3.3 ± 7.1	-0.4 ± 7.5	0.9 ± 7.3	-6.9 ± 8.0

Lat and Long are paleopole coordinates from mean 2 and mean 3 (El Loa Formation mean) of Table 1. A_{95} and K are paleopole 95% confidence and precision parameter [*Fisher*, 1953], N is number of VGPs from Table 1. R (P) are rotation (paleolatitude anomaly). Reference pole A [*Somoza et al.*, 1996] is at 87.4°S, 339.2°E, A_{95}=5.1°; reference pole B [*Aubry et al.*, 1996] is at 84.5°S, 317.4°E, A_{95}=2.3°; Reference pole C is coincident with the present geographic pole using an arbitrary A_{95}=3°.

values of regional rotation are difficult to detect by means of a single paleomagnetic study. Moreover, structural complexities, such as transfer zones, can reduce the magnitude of regional rotation at the expense of higher local values of rotation.

Paleomagnetic results from upper Miocene rocks in the southern Central Andean back arc (Figure 1a) indicate the presence of areas with higher rotation than is predicted by the *Isacks* [1988] differential shortening model (sites "a", "b", and most "d" in Figure 1a). The additional amount of rotation for these localities must be explained by local structures. In at least two of these zones the rotations are related to, or closely associated with, displacement transfer zones in the foreland belt. For instance, along the southeastern margin of the Puna ("d" in Figure 1a), *Aubry et al.* [1996] detected up to 30° of clockwise rotation in upper Miocene rocks within the NE-SW trending Tucumán transfer zone [*Jordan et al.*, 1983; *Allmendinger et al.*,1983; *Allmendinger*, 1986; *de Urreiztieta et al.*, 1996], a broad zone of right-lateral shear produced by transfer of the principal locus of late Miocene-Recent shortening between the Subandean-Santa Bárbara belt to the northeast and the Argentine Precordillera to the southwest.

In the second area associated with a transfer zone the Quebrada Honda section ("a" in Figure 1a) is located directly southwest of, and on strike with, a 20-km-wide transfer zone in the Subandean belt [*Kley*, 1996]. Paleomagnetic data from the section indicate 20° clockwise rotation in 12 Ma flat-lying rocks [*MacFadden et al.*, 1990], which has been used to support models of clockwise rotation of the entire southern Central Andes [*MacFadden et al.*, 1990; 1995; *Butler et al.* 1995; *Watts et al.*, 1995]. *Kley* [1996] describes that early formed structures within the transfer zone are rotated about 15° clockwise in response to differential thrust movement produced by the change in locus of thrusting across the zone (Figure 1b). Although the Quebrada Honda area is located to the west of the zone, removed from the area of displacement transfer, dextral shear and clockwise rotation of structures west of the transfer zone are evident and lead *Kley* [1996] to suggest the development of an E-W trending dextral shear zone (tear fault) in basement rocks involved in the thrusting and active during the transfer zone development in the late Miocene.

MacFadden et al. [1995] report paleomagnetic data from two separate sections in the area of Quehua ("b" in Figure 1a): an upper middle Miocene (around 12.5 Ma) section and an upper Miocene (around 8.6 Ma) one. Integrated results from both the sections suggest 17° clockwise rotation, but there is insufficient information on its structural setting to evaluate the origin of this rotation.

Upper Miocene rocks that do not show paleomagnetic evidence of rotation ("c" and the "star" in Figure 1a) postdate major deformation in their sampling areas [*Somoza et al.*, 1996; this study].

8. Conclusions

Paleomagnetic, rock magnetic, and petrographic data suggest a primary origin for magnetizations isolated in upper Miocene rocks irregularly distributed along a 100-km longitudinal strip in the Precordillera of northern Chile (22°S). Twelve independent records of the paleofield were identified. Paleomagnetic data from two widespread ignimbrites suggest that no relative rotations have occurred between the different sites, and the structural geology of the area suggests that the late Miocene-Recent deformation was not homogeneous. Therefore we conclude the study area has acted as a single coherent structural block with respect to vertical-axis rotation since 8.3 Ma.

Although our paleomagnetic data set may not adequately average paleosecular variation, its time-averaged paleomagnetic direction is indistinguishable from the expected late Miocene reference direction indicating no paleomagnetically significant rotation of the study area since circa 11 Ma. This suggests that any large-scale rigid-body rotation of northern Chile related to late Miocene-Recent crustal shortening in the foreland thrust belt of the Andes must be either very low or nonuniform. Paleomagnetic recognition (if possible) of the former will require a large data set from widely distributed upper Miocene-Pliocene localities in the region as well as a better defined South American apparent polar wander path.

Significantly larger rotations detected in upper Miocene rocks in the southern Central Andes must be of local origin. Available structural data permit us to relate the rotations in at least two localities with dextral shear associated with displacement transfer zones in the late Miocene-Recent foreland thrust system. This and perhaps other yet unexplored mechanisms could account for the tectonic rotations which occurred during the last phase of deformation in the Central Andes.

Several older rotations detected in Mesozoic and Tertiary rocks in the southern Central Andes also have been linked to local structures and explained by distinct mechanisms [*Hartley et al.*, 1992; *Forsythe and Chisholm*, 1994; *Randall et al.*, 1996]. However, the causes for the regional persistence of clockwise rotations, independent of rock unit age, remain poorly understood.

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